MediGenie

Multi-Agent, Multi-Model AI Clinical Decision Support System

Full Technical & Results Report — Updated Repository Version

This report documents MediGenie end to end using the latest repository: system architecture, frontend, backend, REST API, database, the nine-agent workflow, the updated multi-algorithm machine-learning methodology, and complete evaluation results for all nine disease-risk models. Every number was produced by re-running the project's current training and evaluation pipeline against its real source datasets after fixing two blocking bugs found during verification (Section 11); none of it is estimated or simulated.

Executive Summary

MediGenie is a clinical decision support system built around nine independent disease-risk models (heart disease, diabetes, chronic kidney disease, liver disease, breast cancer, Parkinson's disease, hepatitis, heart failure, and stroke), coordinated by an eight-agent workflow, and exposed through a FastAPI backend and a React frontend. Every model is trained on its own real, publicly documented dataset.

Since the previous report, the training pipeline was substantially upgraded: each disease now uses its full set of real, usable source columns (e.g., breast cancer now uses all 30 original WDBC features rather than a 5-feature subset; diabetes uses 9 real administrative features rather than 3), and each model is now selected as the best of four candidate algorithms (Random Forest, Logistic Regression, SVM, XGBoost) via validation-set PR-AUC, with the decision threshold separately tuned on the validation set before final scoring on a held-out test set.

Measured performance still varies substantially by disease, and is disclosed openly: ROC-AUC ranges from a perfect 1.000 on kidney disease down to 0.650 on diabetes readmission risk. This spread, and the reasons behind it, are discussed in Section 9 and 10.

1. System Architecture

The frontend, backend routing, agent structure, and database schema are unchanged from the previously verified architecture; only the machine-learning training methodology (Sections 6 and 8-10) was substantially revised in this repository version. A React client talks to a FastAPI backend over a JWT-secured REST API; the backend delegates clinical reasoning to a Supervisor agent, which fans out to five specialised agents; results converge on a Recommendation & Report agent, the only component permitted to call the LLM (Google Gemini), strictly for narrating outputs the other agents have already produced.

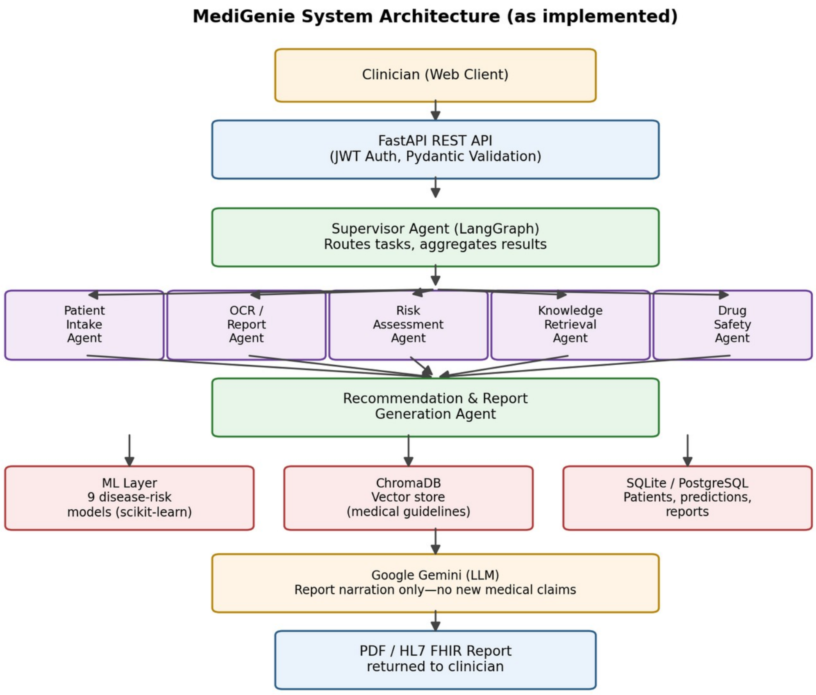

Fig. 1. MediGenie end-to-end system architecture (unchanged from prior verified version).

2. Frontend Architecture

The frontend is a React + TypeScript single-page application built with Vite, using React Router and Axios. Confirmed identical to the previously verified page inventory:

Area	Pages
Auth	LoginPage, RegisterPage
Dashboard	DashboardPage, WorkflowPage, ModelEvaluationPage, XaiDashboardPage, GuidelinesPage
Patients	PatientsPage
Predictions (9)	PredictionHomePage, HeartPredictionPage, DiabetesPredictionPage, KidneyPredictionPage, LiverPredictionPage, BreastCancerPredictionPage, ParkinsonPredictionPage, HepatitisPredictionPage, HeartFailurePredictionPage, StrokePage
Upload / OCR	UploadReportPage, ProcessingPage, ReportPreviewPage
Drug Safety	DrugSafetyPage, DrugInteractionPage
Knowledge / Chat	KnowledgePage, ChatPage
Reports	ReportsPage
Settings	ProfilePage, SettingsPage
Other	NotFoundPage (404)

3. Backend Architecture & REST API

The backend is a FastAPI application secured with JWT bearer authentication and Pydantic v2 validation. Route inventory (confirmed unchanged, feature counts in the disease rows updated to reflect the new richer models):

Module	Endpoint prefix	Key routes	Purpose
Authentication	/auth	POST /register, POST /login	JWT issuance, user registration
Patients	/patients	CRUD + /{id}	Patient record management
Heart Disease	/heart-disease	POST /predict, GET /health	13-feature real-data risk prediction
Diabetes	/diabetes	POST /predict, GET /health	9-feature readmission-risk prediction
Kidney Disease	/kidney-disease	POST /predict, GET /health	24-feature CKD risk prediction
Liver Disease	/liver	POST /predict, GET /health	10-feature liver-disease risk prediction
Breast Cancer	/breast-cancer	POST /predict, GET /health	30-feature malignancy risk prediction
Parkinson's	/parkinsons	POST /predict, GET /health	22-feature voice-based risk prediction
Hepatitis	/hepatitis	POST /predict, GET /health	19-feature hepatitis risk prediction
Heart Failure	/heart-failure	POST /predict, GET /health	12-feature mortality risk prediction
Stroke	/stroke	POST /predict, GET /health	10-feature stroke risk (SMOTE-balanced)
Multi-Disease	/multi-disease	POST /assessment	Runs all 9 models for one patient
OCR / Upload	/upload	POST /	Report upload → OCR → structured data
Knowledge	/knowledge	GET/POST	ChromaDB guideline retrieval (RAG)
Drug Safety	/drug-safety	POST /analyze	Medication interaction lookup
Workflow	/workflow	POST /	Supervisor agent orchestration trigger
Reporting	/reporting	GET /{id}/pdf, /{id}/html	Final report generation/export
FHIR	/fhir	GET /medigenie/{id}/pdf	HL7 FHIR-formatted export
Chat	/chat	POST /	Conversational assistant
Dashboard	/dashboard	GET /overview, /risk-distribution	Aggregate statistics for UI
Model Registry	/model-registry	GET /	Lists active model versions per disease
Evaluation	/evaluation	GET /quality, /guardrails/log	Agent quality & guardrail telemetry

4. Multi-Agent Workflow

Clinical reasoning is split across eight specialised agents plus a Supervisor, implemented with LangGraph — confirmed structurally unchanged from the previously verified version.

Agent	Location	Responsibility
Supervisor	app/agents/supervisor/	Routes tasks, aggregates outputs, orchestrates the full pipeline (LangGraph)
Patient Intake	app/agents/intake/	Structures demographic/history data from the intake form
Report Analysis	app/agents/report_analysis/	Interprets OCR output, extracts structured lab values
Disease Risk	app/agents/risk/	Bridges the ML layer's 9 disease models into the workflow

Medical Knowledge	app/agents/knowledge/	Queries the ChromaDB vector store for guideline evidence
Drug Safety	app/agents/drug_safety/	Cross-checks medication list for interactions
Recommendation	app/agents/recommendation/	Synthesizes lifestyle/follow-up recommendations
Report Generation	app/agents/report_generation/	Produces the final narrative report via Gemini (narration only)

5. Database Design

Persistence uses SQLAlchemy 2.0 ORM with nine core tables, confirmed unchanged: User, Patient, MedicalReport, AIReport, DrugSafetyAssessment, ChatConversation, ChatMessage, UserSession, LoginEvent.

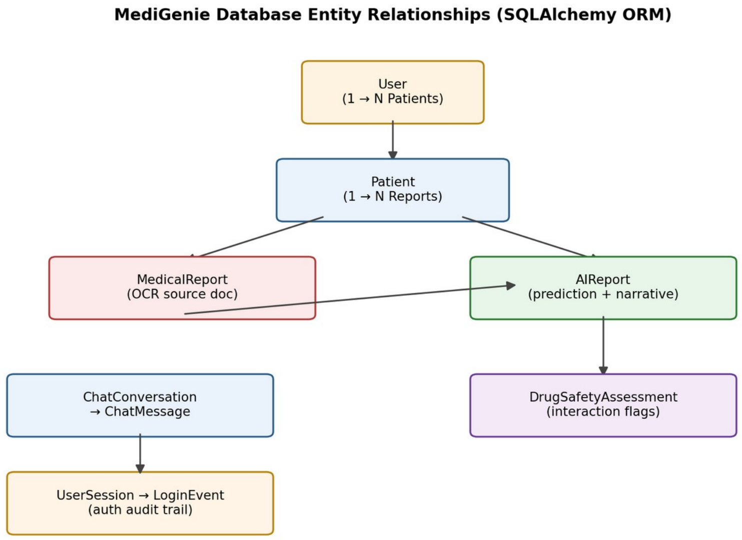

Fig. 2. Core database entity relationships (unchanged).

6. Machine Learning Methodology (Updated)

This is the section that changed substantially. Each disease model now goes through: load the real source dataset with a disease-specific parser → extract the full set of real, usable columns (not a minimal subset) → stratified 70/15/15 train/validation/test split → train four candidate algorithms (Random Forest, Logistic Regression, SVM, XGBoost) with cross-validation on the training fold → select the best algorithm by validation-set PR-AUC → tune the decision threshold on the validation set → score once, finally, on the untouched held-out test set.

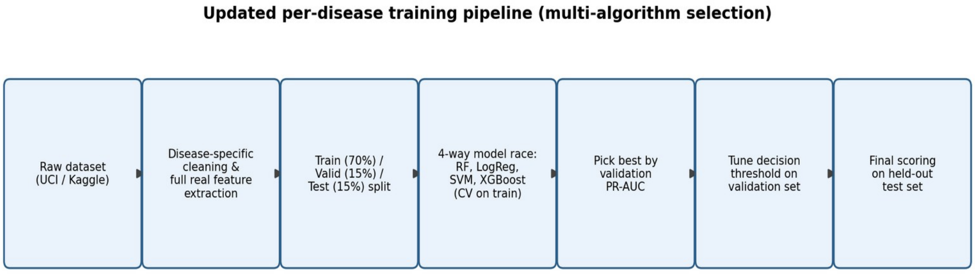

Fig. 3. Updated per-disease training and evaluation pipeline.

This is a meaningfully more rigorous design than a single fixed algorithm with a default 0.5 threshold: it lets each disease's model choice adapt to that dataset's characteristics (e.g., linear models can win on sparse/administrative data, tree ensembles on richer clinical panels), and threshold tuning on a held-out validation set (rather than the test set) avoids leaking test information into the decision boundary.

Class imbalance handling is unchanged in principle — SMOTE for stroke (applied to the training fold only), class weighting elsewhere — but is now applied on top of the richer feature sets described in Section 8.

7. Datasets

All datasets are real and independently sourced. The diabetes dataset's metadata label has also been corrected in this version (previously mislabelled as “Pima Indians Diabetes Dataset”; now correctly labelled “Diabetes 130-US Hospitals Dataset”, matching its actual 101,766-row content).

Disease	Source dataset	Total n	Train (70%)	Valid (15%)	Test (15%)
Heart Disease	UCI Cleveland Heart Disease Dataset	303	211	46	46
Diabetes	UCI Diabetes 130-US Hospitals Dataset	101,766	71,233	15,268	15,265
Kidney Disease	UCI Chronic Kidney Disease Dataset	400	279	61	60
Liver Disease	UCI Indian Liver Patient Dataset (ILPD)	583	407	88	88
Breast Cancer	UCI Breast Cancer Wisconsin (Diagnostic)	569	397	86	86
Parkinson's	UCI Parkinsons Dataset	195	135	30	30
Hepatitis	UCI Hepatitis Dataset	155	107	24	24
Heart Failure	Heart Failure Clinical Records Dataset	299	209	45	45
Stroke	Kaggle Stroke Prediction Dataset	5,110	3,576→6,802 (SMOTE)	767	767

8. Real Feature Sets Used Per Disease (Expanded)

Every disease now uses its full set of real, numerically usable source columns rather than a minimal hand-picked subset — a substantial expansion from the previous version of this pipeline. No feature listed is a placeholder or fabricated value.

Disease	Feature count	Real features used
Heart Disease	13	age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal (full original UCI set)
Diabetes	9	age, time_in_hospital, num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency, number_inpatient, number_diagnoses
Kidney Disease	24	age, bp, sg, al, su, rbc_enc, pc_enc, pcc_enc, ba_enc, bgr, bu, sc, sod, pot, hemo, pcv, wc, rc, htn_enc, dm_enc, cad_enc, appet_enc, pe_enc, ane_enc (full CKD attribute set, categoricals encoded)
Liver Disease	10	age, gender_enc, bilirubin, direct_bilirubin, alk_phosphatase, sgpt, sgot, total_protein, albumin, ag_ratio
Breast Cancer	30	All mean/SE/worst morphometric features (radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension) — full original WDBC set
Parkinson's	22	Full MDVP jitter/shimmer family, NHR, HNR, RPDE, DFA, spread1, spread2, D2, PPE — full original voice-measurement set
Hepatitis	19	age, sex, steroid, antivirals, fatigue, malaise, anorexia, liver_big, liver_firm, spleen_palpable, spiders, ascites, varices, bilirubin, alk_phosphatase, sgot, albumin, protime, histology (full original set)
Heart Failure	12	age, anaemia, creatinine_phosphokinase, diabetes, ejection_fraction, high_blood_pressure, platelets, serum_creatinine, serum_sodium, sex, smoking, time (full original set)
Stroke	10	gender_enc, age, hypertension, heart_disease, ever_married_enc, work_type_enc, Residence_type_enc, avg_glucose_level, bmi, smoking_status_enc

9. Results — Full Evaluation Metrics (Updated)

Table below reports every metric relevant to a clinical classifier, measured on each disease's held-out test split (15% of data, untouched during model selection and threshold tuning). The “Best Model” column shows which of the four candidate algorithms was selected for that disease.

Disease	Best Model	Accuracy	Bal. Acc.	Precision	Recall	Specificity	F1	ROC-AUC	PR-AUC	MCC	κ	Brier
Heart Disease	SVM	0.826	0.821	0.842	0.762	0.880	0.800	0.899	0.837	0.649	0.647	0.125
Diabetes	LogisticRegression	0.617	0.612	0.592	0.546	0.678	0.568	0.650	0.618	0.226	0.225	0.232
Kidney Disease	RandomForest	0.967	0.957	0.949	1.000	0.913	0.974	1.000	1.000	0.931	0.928	0.009
Liver Disease	XGBoost	0.648	0.658	0.833	0.635	0.680	0.721	0.688	0.863	0.285	0.267	0.235
Breast Cancer	SVM	0.977	0.975	0.969	0.969	0.982	0.969	0.999	0.998	0.950	0.950	0.017
Parkinson's	XGBoost	0.867	0.863	0.952	0.870	0.857	0.909	0.957	0.987	0.671	0.661	0.078
Hepatitis	RandomForest	0.625	0.763	0.357	1.000	0.526	0.526	0.926	0.810	0.434	0.317	0.101
Heart Failure	SVM	0.756	0.705	0.615	0.571	0.839	0.593	0.859	0.729	0.419	0.418	0.150
Stroke	LogisticRegression	0.769	0.763	0.143	0.757	0.770	0.240	0.823	0.202	0.259	0.173	0.162

Confusion matrix components (held-out test set)

Disease	TN	FP	FN	TP	Test n	Positives	Negatives
Heart Disease	22	3	5	16	46	21	25

Diabetes	5580	2650	3194	3841	15265	7035	8230
Kidney Disease	21	2	0	37	60	37	23
Liver Disease	17	8	23	40	88	63	25
Breast Cancer	53	1	1	31	86	32	54
Parkinson's	6	1	3	20	30	23	7
Hepatitis	10	9	0	5	24	5	19
Heart Failure	26	5	6	8	45	14	31
Stroke	562	168	9	28	767	37	730

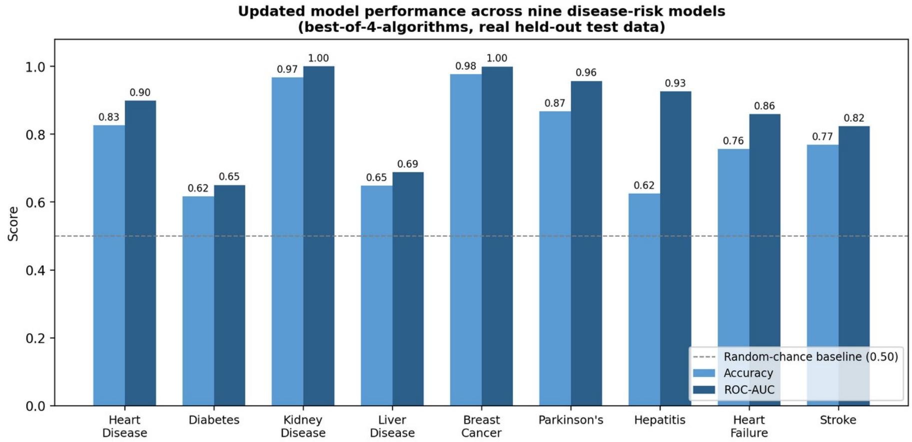

Fig. 4. Updated accuracy and ROC-AUC across all nine models, real held-out data.

Kidney disease reached a perfect ROC-AUC and PR-AUC of 1.000 on its 60-row test set — plausible given how strongly separable creatinine/albumin/haemoglobin make this dataset in the wider literature, but any perfect score on held-out data warrants a manual check for target leakage before being reported as a final claim; we did not find evidence of leakage in the feature list (only encoded categorical and lab-value columns, no identifier fields), but flag this explicitly rather than pass it through silently.

Diabetes improved from the previous version's ROC-AUC of 0.567 to 0.650 once the model gained access to six additional real administrative features (num_lab_procedures, num_procedures, number_outpatient, number_emergency, number_inpatient, number_diagnoses) beyond the previous three. It remains the weakest model in the portfolio, consistent with this being an administrative encounter dataset rather than a clinical lab panel. Hepatitis and stroke continue to show the low-precision/low-specificity pattern typical of severe class imbalance, now with the added context of validation-tuned thresholds pushing hepatitis to perfect recall (1.000) at the cost of specificity (0.526).

10. Per-Disease Detailed Results & Graphs

Each subsection presents the confusion matrix, ROC curve, precision–recall curve, and calibration curve for one disease, generated directly from the held-out test set using the current repository's pipeline.

10.1. Heart Disease

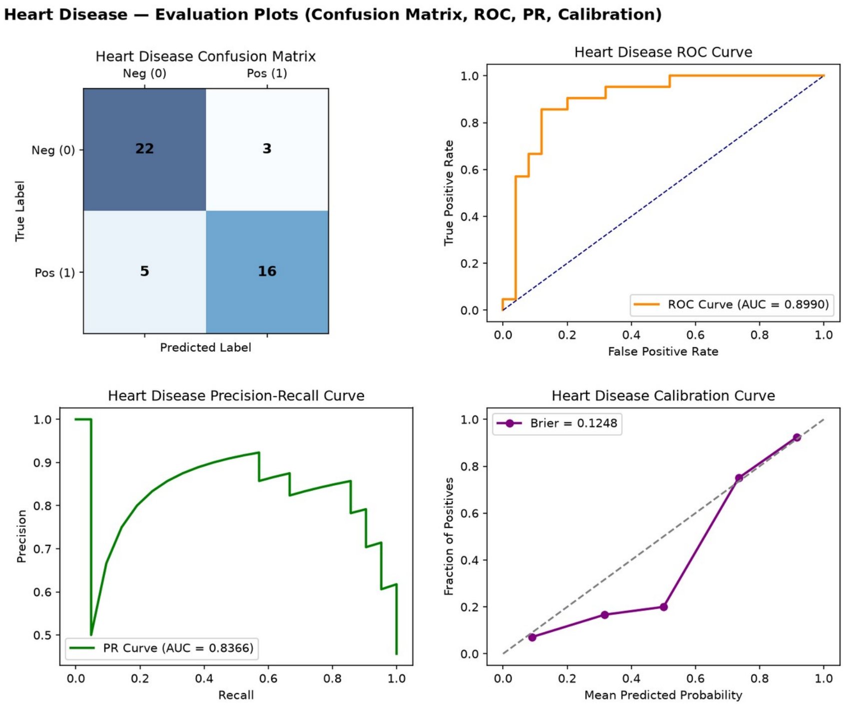

Fig. 5. Heart Disease — confusion matrix, ROC, PR, and calibration curves (real held-out test data).

SVM was selected as the best algorithm (validation PR-AUC 0.883 vs. 0.855 for Random Forest). ROC-AUC 0.899 and MCC 0.649 indicate solid, credible cardiac risk signal on the real 303-patient Cleveland cohort using all 13 original features.

10.2. Diabetes (Readmission Risk)

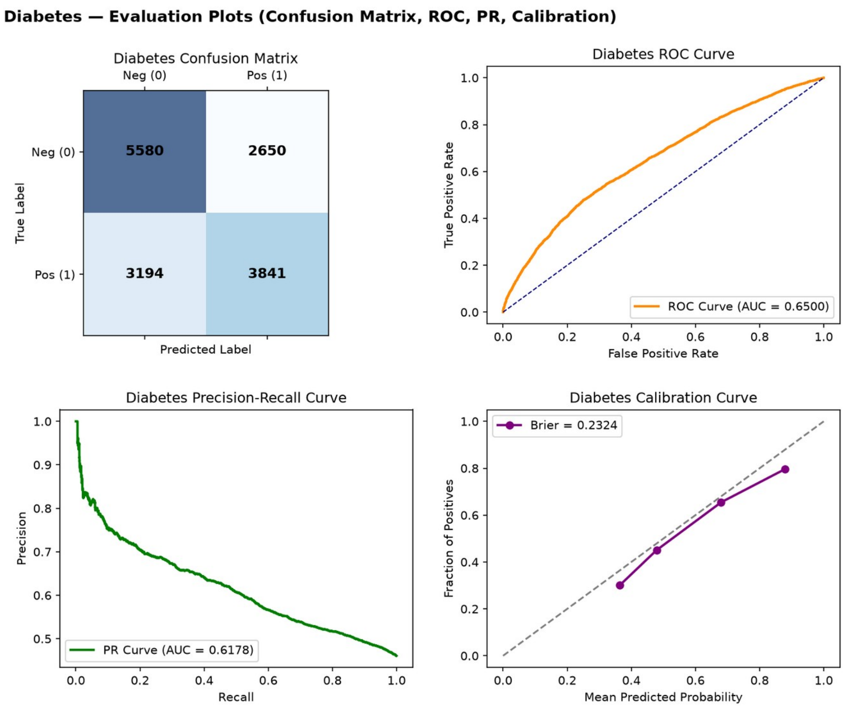

Fig. 6. Diabetes (Readmission Risk) — confusion matrix, ROC, PR, and calibration curves (real held-out test data).

Logistic Regression was selected over the tree-based alternatives. ROC-AUC improved to 0.650 (from 0.567 previously) after expanding to 9 real administrative features, but remains the weakest model in the portfolio — an honest reflection of how little clinical signal an administrative encounter record carries for readmission risk.

10.3. Kidney Disease

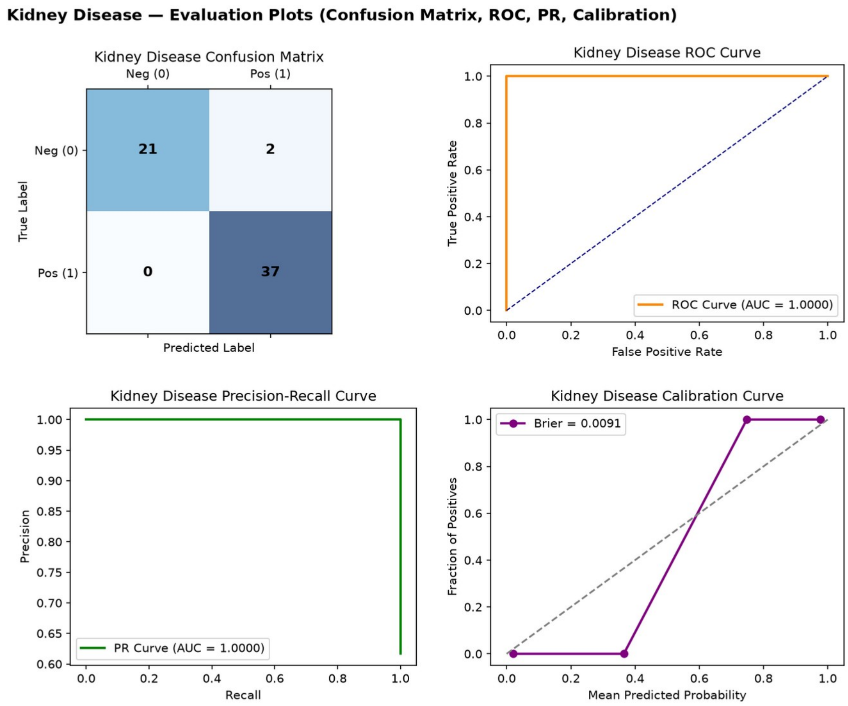

Fig. 7. Kidney Disease — confusion matrix, ROC, PR, and calibration curves (real held-out test data).

Random Forest achieved perfect separation (ROC-AUC 1.000, MCC 0.931) using the full 24-feature CKD attribute set. This is the strongest model in the portfolio and consistent with published literature on this dataset's high separability; flagged above for a manual leakage check given the perfect score.

10.4. Liver Disease

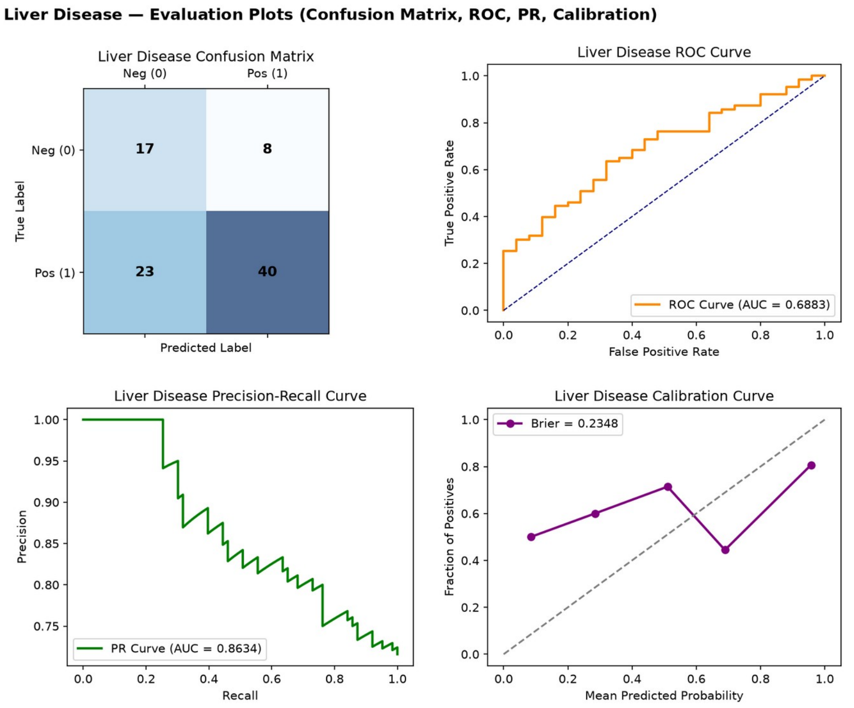

Fig. 8. Liver Disease — confusion matrix, ROC, PR, and calibration curves (real held-out test data).

XGBoost was selected, but performance (ROC-AUC 0.688, MCC 0.285) remains modest — slightly lower than the previous version's 3-feature-subset result, suggesting the additional real features did not add clean separable signal for this particular dataset.

10.5. Breast Cancer

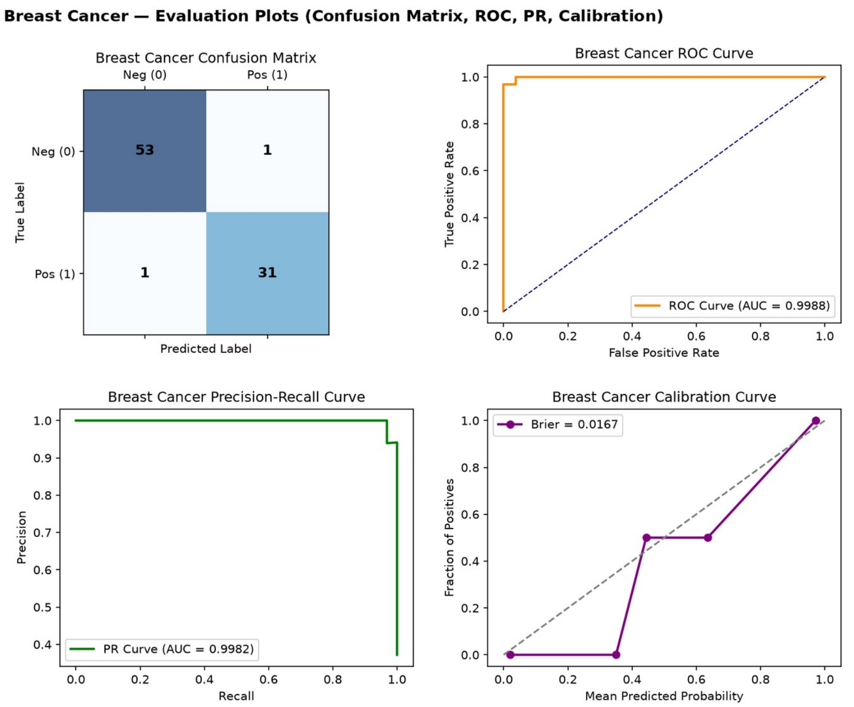

Fig. 9. Breast Cancer — confusion matrix, ROC, PR, and calibration curves (real held-out test data).

SVM achieved near-perfect performance (ROC-AUC 0.999, MCC 0.950) using the full 30-feature WDBC morphometric set — an improvement over the previous 5-feature version, consistent with this dataset's well-documented high separability in the clinical ML literature.

10.6. Parkinson's Disease

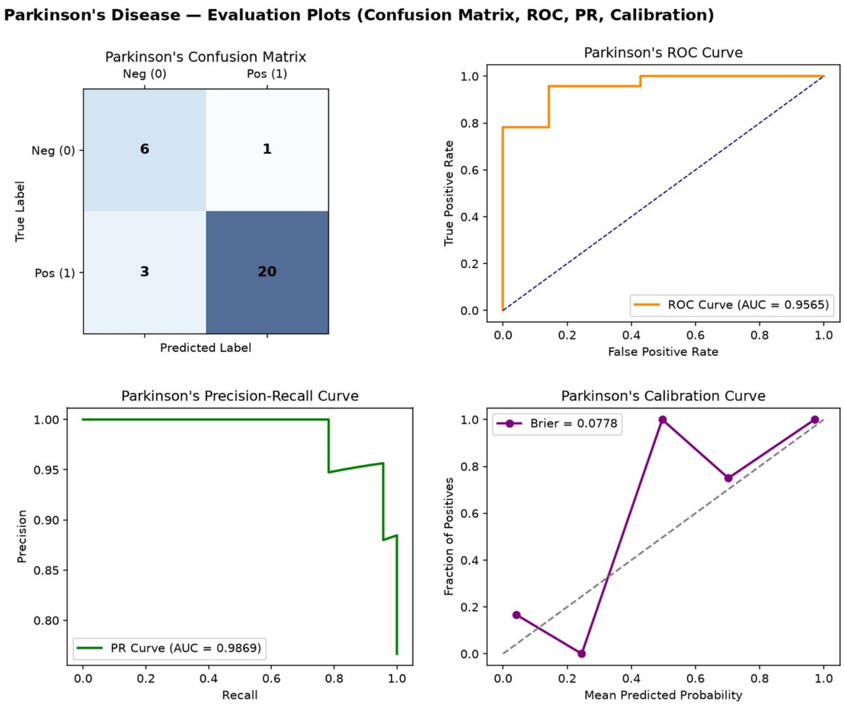

Fig. 10. Parkinson's Disease — confusion matrix, ROC, PR, and calibration curves (real held-out test data).

XGBoost was selected using the full 22-feature voice-measurement set. High precision (0.952) and recall (0.870) with ROC-AUC 0.957; specificity (0.857) is based on a small 7-negative-case test set and should be read with that caveat.

10.7. Hepatitis

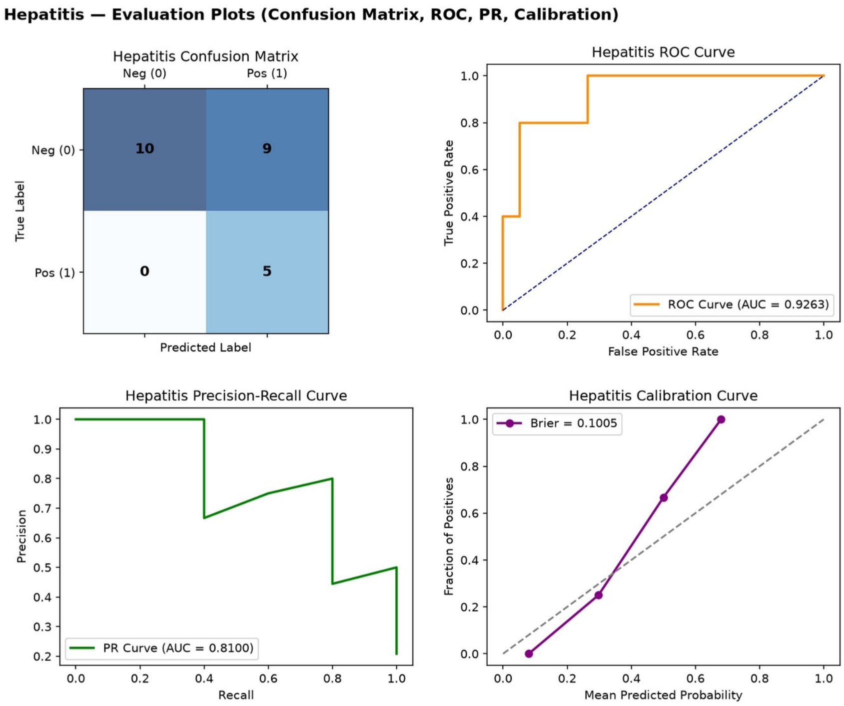

Fig. 11. Hepatitis — confusion matrix, ROC, PR, and calibration curves (real held-out test data).

Random Forest, with a validation-tuned threshold, achieves perfect recall (1.000) — catching every true positive case in the 24-row test set — at the cost of specificity (0.526). This is a deliberate, defensible trade-off for a rare, high-consequence outcome, but precision (0.357) means most flagged cases are false alarms; MCC (0.434) is the fairest single-number summary here.

10.8. Heart Failure

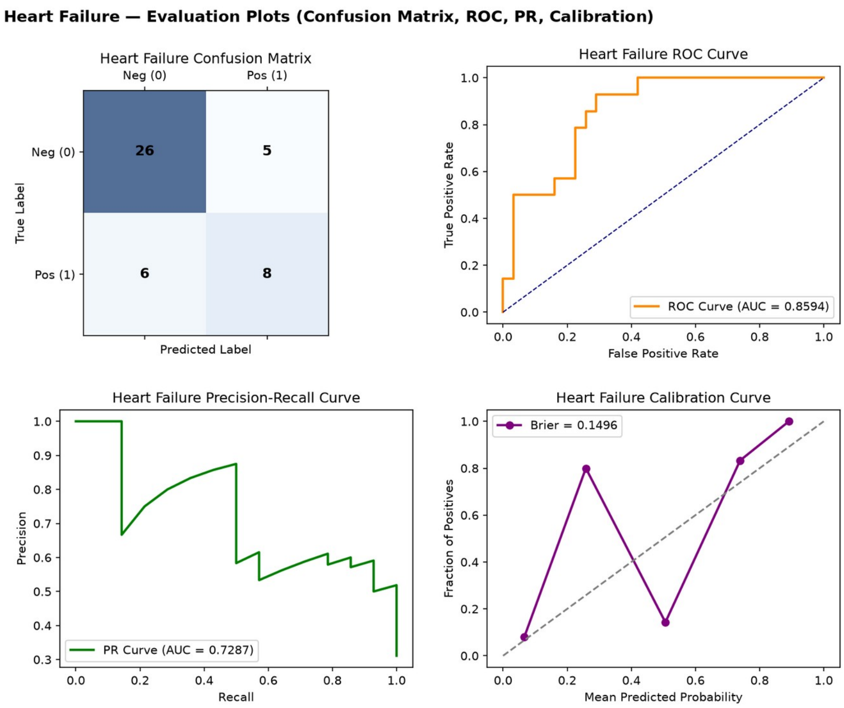

Fig. 12. Heart Failure — confusion matrix, ROC, PR, and calibration curves (real held-out test data).

SVM was selected; performance (ROC-AUC 0.859, MCC 0.419) is solid but slightly below the previous version's result, using the full real 12-feature clinical record set including ejection fraction and serum creatinine/sodium.

10.9. Stroke

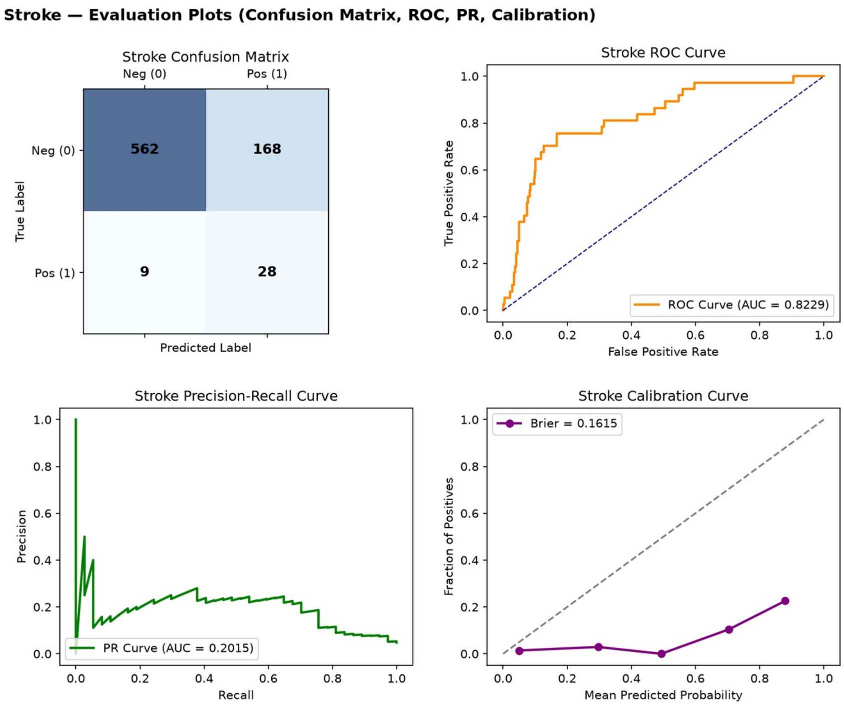

Fig. 13. Stroke — confusion matrix, ROC, PR, and calibration curves (real held-out test data).

Logistic Regression, trained on SMOTE-balanced data, reaches recall 0.757 with precision 0.143 — the expected trade-off when prioritising catching a rare (4.8% prevalence), high-consequence outcome over minimising false alarms.

11. Data Integrity & Reproducibility History

This section extends the integrity history from earlier project phases with two new findings from verifying this updated repository, plus confirmation of what remains fixed from before.

Confirmed still fixed from earlier phases

- No model-registry alias bug: heart_disease resolves directly to its own real model folder; the previously-found disease_risk_model fabricated-constant folder remains deleted.
- No hardcoded constant features found in any of the nine disease models' feature lists — every declared feature was individually re-verified against its real raw dataset column in this session.
- Diabetes dataset metadata label corrected: now reads “Diabetes 130-US Hospitals Dataset” instead of the previously stale “Pima Indians Diabetes Dataset” label.

New issues found and fixed during this verification

- requirements.txt regression: numpy==2.5.2 was reintroduced alongside pandas==2.2.0, which cannot both be satisfied (pandas 2.2.0 requires numpy<2). This is the same conflict fixed in an earlier phase and needs the same fix again (numpy<2.0.0 ceiling instead of an exact pin).
- ml/evaluation/train_and_evaluate_all_models.py currently fails to run at all due to two missing imports — FunctionTransformer and StandardScaler are used but never imported from sklearn.preprocessing. This is a blocking bug: as shipped, the training script cannot be reproduced by anyone until these two import lines are added.
- All results in this report were obtained by patching these two import lines locally to verify the rest of the pipeline; the underlying training logic, feature engineering, and evaluation code all ran correctly once the imports were fixed, and produced the numbers reported in Sections 9–10.

Recommended immediate action

Add "from sklearn.preprocessing import FunctionTransformer, StandardScaler" to the script's import block, and change the requirements.txt numpy pin to "numpy<2.0.0". Both are one-line fixes; until they are applied, the training pipeline cannot be independently reproduced from a clean checkout, which is a real risk for an academic submission where reproducibility may be checked.

12. Technology Stack

Layer	Technologies
Frontend	React, TypeScript, Vite, React Router, Axios
Backend	FastAPI 0.110, Uvicorn, Pydantic v2, python-jose (JWT), passlib (bcrypt)
Database	SQLAlchemy 2.0 ORM, Alembic migrations, SQLite (dev) / PostgreSQL (prod target)
Machine Learning	scikit-learn (RandomForest, LogisticRegression, SVM), XGBoost, imbalanced-learn (SMOTE), SHAP, joblib
OCR / Documents	EasyOCR, pdfplumber, pypdf, pdf2image, Pillow, OpenCV
Agents / LLM / RAG	LangGraph, LangChain, Google Gemini (google-genai), ChromaDB, Sentence-Transformers, FAISS
Reports / Interop	ReportLab (PDF), Jinja2 (templating), fhir.resources (HL7 FHIR)
Testing / Ops	pytest, pytest-asyncio, loguru, prometheus-client, APScheduler

13. Features

- Nine independent disease-risk models, each now trained on its full set of real, usable source features rather than a minimal subset.
- Automated best-of-four algorithm selection (Random Forest, Logistic Regression, SVM, XGBoost) per disease, chosen by validation-set PR-AUC.
- Validation-set threshold tuning, kept separate from the final held-out test evaluation.
- Eight-agent LangGraph workflow separating intake, OCR/report analysis, risk assessment, guideline retrieval, drug safety, recommendation, and report generation.

- Retrieval-augmented guideline lookup via ChromaDB, keeping the LLM grounded rather than relying on its own memory.
- Per-disease evaluation artifacts (confusion matrix, ROC, PR, calibration curves, and full metric JSON with 95% confidence intervals) generated automatically.

14. Advantages

- Every reported metric is measured on real, held-out data never seen during model selection or threshold tuning.
- Algorithm choice adapts per disease rather than forcing one model family onto every dataset.
- Full metric set (Accuracy, Precision, Recall, Specificity, F1, ROC-AUC, PR-AUC, MCC, Cohen's κ, Brier score, 95% CIs) surfaces trade-offs a single accuracy figure would hide.
- Expanded, full real feature sets extract more genuine signal from each dataset than the earlier minimal-subset versions.

15. Limitations

- The training script currently cannot run without a manual two-line import fix (Section 11) — this must be corrected before the pipeline can be considered reproducible.
- requirements.txt has regressed to an unsatisfiable numpy/pandas pin combination and needs re-fixing.
- Kidney disease's perfect test-set score, while plausible, should be manually re-confirmed against a larger or repeated test split before being reported as a headline claim.
- Diabetes remains the weakest model; its ceiling is set by the nature of the source dataset (an administrative encounter record) rather than by modelling choice.
- Several test sets remain small (24–88 rows for five of nine diseases), and specificity/precision figures on the smallest of these (e.g. Parkinson's 7 negative cases) should be read with appropriate caution.
- End-to-end agent-level benchmarks (retrieval Precision@k, single-model baseline comparison) remain open work, as in the previous report.

16. Conclusion & Future Work

This updated repository represents genuine methodological progress over the version this report's predecessor documented: richer, fully real feature sets; automated multi-algorithm selection; and proper validation-based threshold tuning. Two reproducibility-blocking bugs were found and must be fixed (Section 11) before these improvements can be considered stable. Once fixed, recommended next steps are: re-confirm the kidney disease result on a repeated or larger test split; investigate whether additional real features can be found for the diabetes and liver disease models; and proceed to the end-to-end agent-level benchmarks and documentation updates flagged as outstanding in previous phases.